

In the original analysis of the fusion-time data, the experimenters concluded that there was no support for the hypothesis that prior information about the object reduces fusion time. Yet our analysis here does seem to suggest an effect: the VV distribution is shifted toward lower values by a factor of $2/3$. This apparent conflict between the original analysis and the analysis here is discussed in the final section of this chapter where it is argued that our visualization of the data has yielded valid insight that was missed by the experimenters.

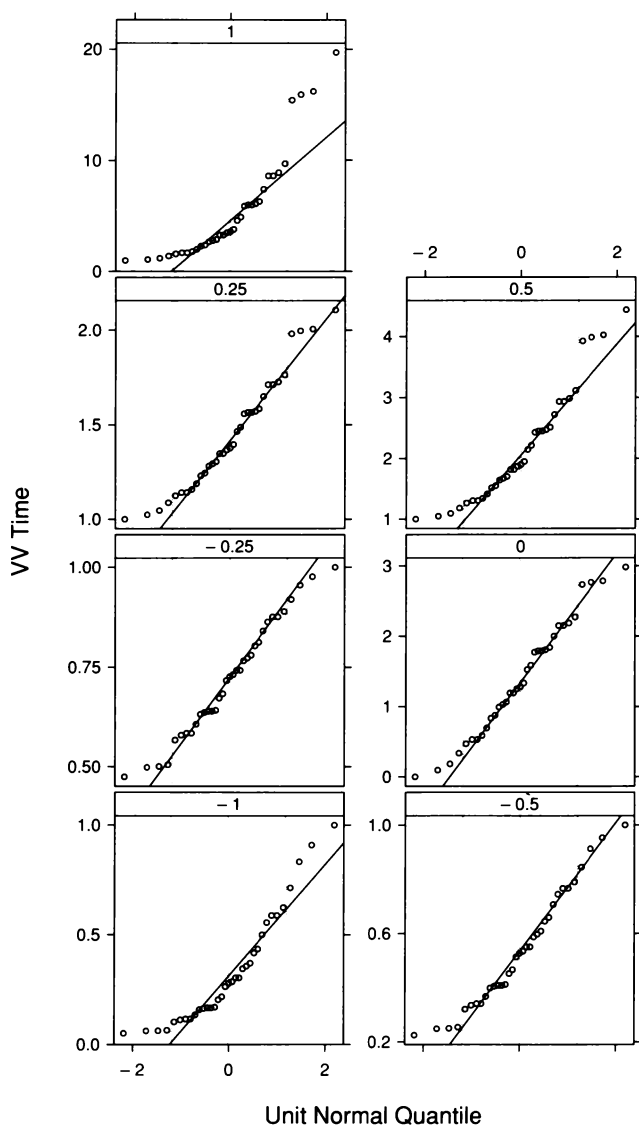
2.7 Power Transformation

We need more than just the logarithm in our arsenal of transformation methods. Logs cannot be used for data with zeros unless the data are adjusted. And a transformation other than the logarithm is often the one that leads to a simple data structure; in many cases the logarithm can fail to cure skewness and monotone spread, but another transformation does so.

Power transformations are a class of transformations that includes the logarithm. Let x be the variable under study. The power transformation with parameter τ is defined to be x^τ if $\tau \neq 0$ and $\log(x)$ if $\tau = 0$. For $\tau = 1$, the transformation is just x , so this leaves the data alone. For $\tau = 1/2$, the transformation is the square root, $\sqrt{x}$. For $\tau = -1$, it is the inverse of the data, $1/x$. It might seem artificial to define the power transformation for $\tau = 0$ to be the logarithm, but in fact it belongs there because x^τ for τ close to zero behaves much like the logarithm; for example, the derivative of $\log x$ is x^{-1} , and the derivative of $x^{0.001}$ is proportional to $x^{-.999}$. The parameter τ can be any number if the data are positive, but τ must be greater than 0 if the data have zeros. Of course, if the data have negative values, no power transformation is possible without some adjustment of the data.

Figure 2.33 shows normal q-q plots of the VV fusion times transformed by seven power transformations with values of τ equal to -1 , $-1/2$, $-1/4$, 0 , $1/4$, $1/2$, and 1 . The panels are in graphical order: the value of τ increases as we go from left to right and from bottom to top through the panels. The figure illustrates a phenomenon that occurs for many data sets that are skewed toward large values. As τ decreases from 1, the skewness is reduced until the data become nearly

symmetric, and then as τ is further reduced, the data become more and more skewed again. When τ goes from 1 to $1/2$, the transformation pushes the upper tail closer to the center and pulls the lower tail further from the center. This continues as τ decreases, although when τ goes negative, the transformation reverses the upper and lower tails, so the continued force of the transformation pulls out the upper tail and pushes in the lower.



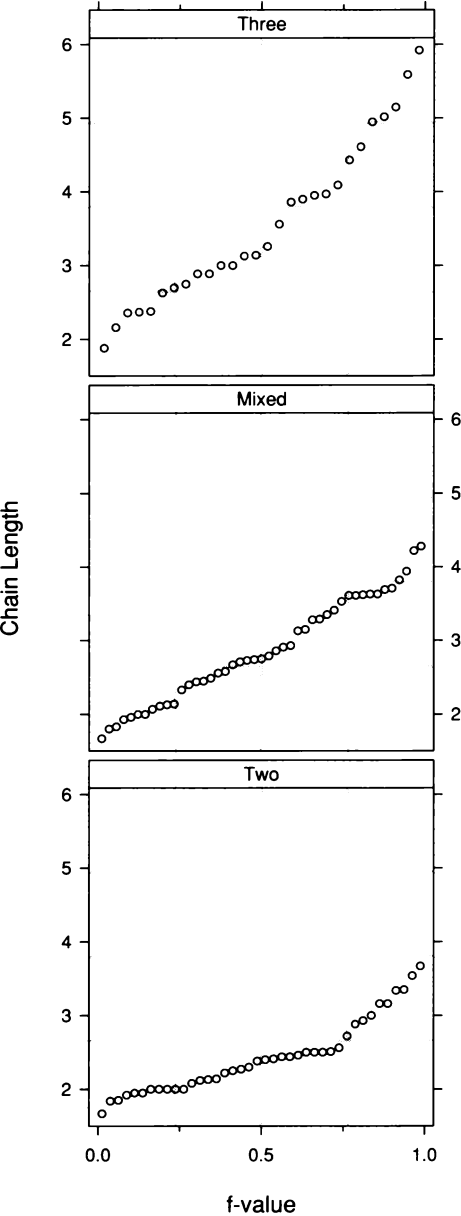
2.33 Seven power transformations of the VV times are displayed by normal q-q plots.

Figure 2.33 illustrates the method for discovering the power transformation that brings a distribution the closest to symmetry — trial and error. We simply choose a selection of values of τ , graph the transformed data for each transformation, and assess the symmetry. The seven values of τ in Figure 2.33 are a good starting set because they provide a representative collection of power transformations. For the VV times, the logarithm and the inverse fourth root do the best job. They also do the best job for the NV times. But, of course, a tie goes to the logarithm because of the multiplicative interpretation.

Food Webs

The food web for the animal species in an ecosystem is a description of who eats whom. A chain is a path through the web. It begins with a species that is eaten by no other, moves to a species that the first species eats, moves next to a species that the second species eats, and so forth until the chain ends at a species that preys on no other. If there are 7 species in the chain then there are 6 links between species, and the length of the chain is 6. The mean chain length of a web is the mean of the lengths of all chains in the web.

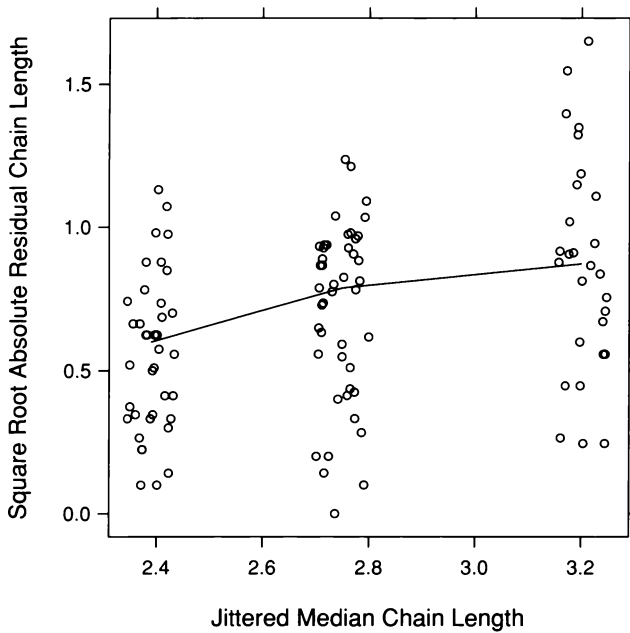
A two-dimensional ecosystem lies in a flat environment such as a lake bottom or a grassland; movement of species in a third dimension is limited. In a three-dimensional ecosystem, there is considerable movement in three dimensions. One example is a forest canopy; another is a water column in an ocean or lake. A mixed ecosystem is made up of a two-dimensional environment and a three-dimensional environment with enough links between the two to regard it as a single ecosystem. An interesting study reports the mean chain lengths for 113 webs [11]. Quantile plots display the data in Figure 2.34. Here, we will study how the distributions of mean chain lengths vary for the three classes — two-dimensional, mixed, and three-dimensional. In doing this, we regard the dimensionality of the mixed webs as lying between that of the two-dimensional webs and the three-dimensional webs.



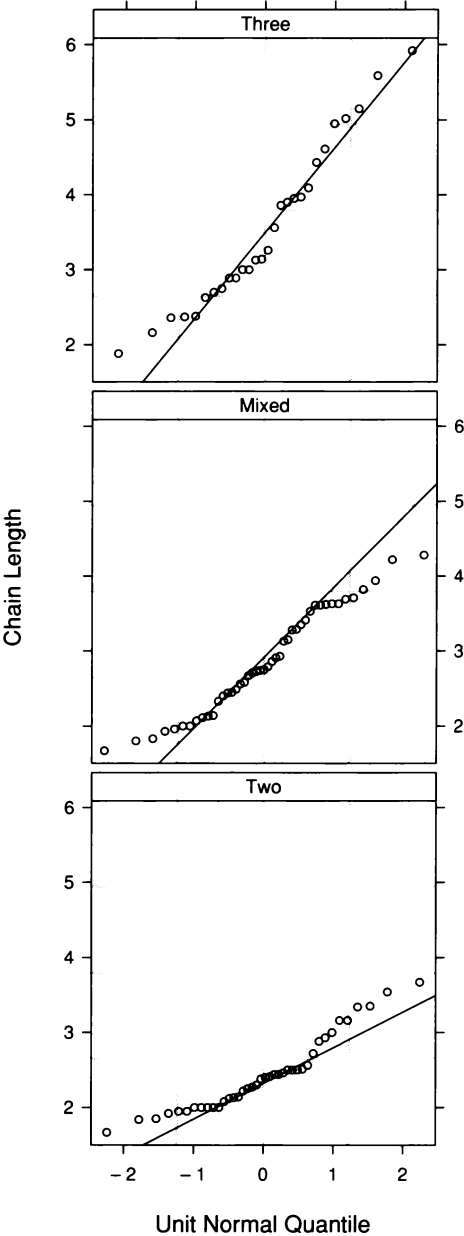
2.34 Quantile plots display the chain length measurements for three ecosystem dimensions.

Skewness and Monotone Spread

Figure 2.35 is an s-l plot of the three distributions of chain length. There is monotone spread. Normal q-q plots in Figure 2.36 reveal mild skewness toward large values. Also, the middle panel shows a peculiarity in the upper tail of the webs of mixed dimension. At a length of about 3.5, there is a knee in the distribution caused by five webs with very nearly identical values.

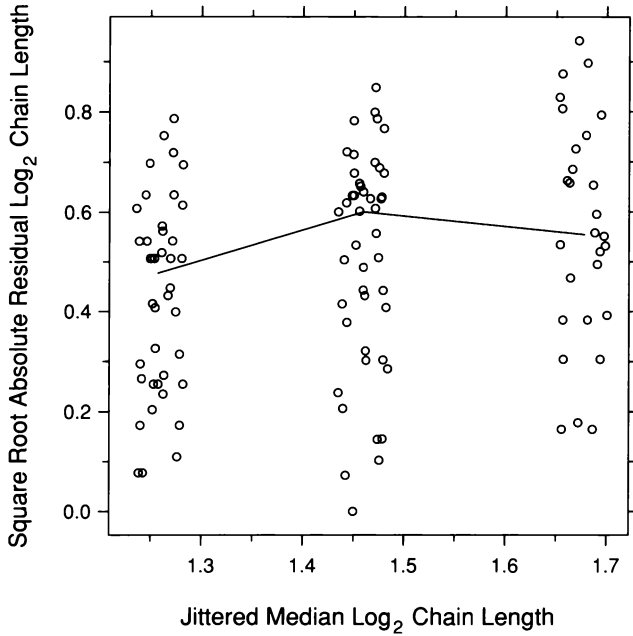


2.35 The s-l plot for the chain lengths checks for nonuniform spread.

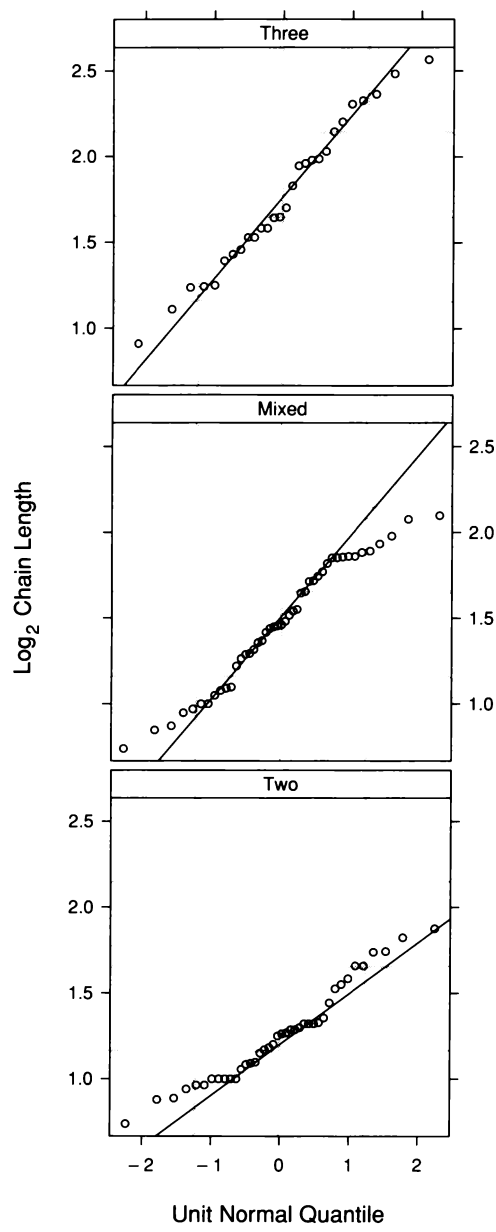


2.36 Normal q-q plots compare the chain length distributions with the normal distribution.

The skewness and monotone spread of the food web data are not cured by the logarithm, although the transformation does reduce their severity. Figure 2.37 is an s-l plot for the logs. The monotone spread remains, although the magnitude has been substantially reduced. Figure 2.38 shows normal q-q plots. Skewness remains, particularly for dimension two. We need a smaller value of τ .



2.37 The s-l plot for the log chain lengths checks for nonuniform spread.



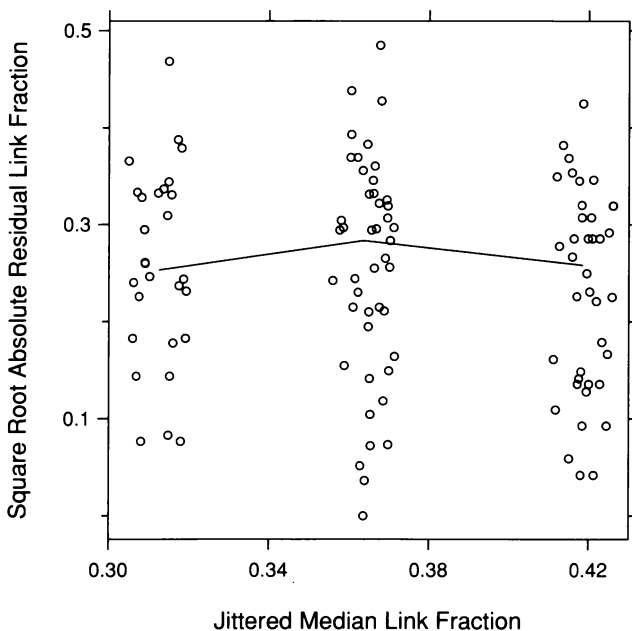
2.38 Normal q-q plots compare the log chain length distributions with the normal distribution.

The inverse transformation, which is $\tau = -1$, does the best job although the improvement over the logarithm is small. Figure 2.39 is the s-l plot for the inverse lengths; monotone spread no longer occurs. Figure 2.40 shows normal q-q plots of the inverses. The peculiar behavior for the mixed dimension now occurs in the lower tail because the inverse transformation has changed the order of the measurements. The other two panels, however, show a small amount of convexity in the lower tail of the distribution, so some of the peculiar behavior for the mixed dimension appears to be part of a general pattern.

The inverse transformation provides a natural measurement scale for the food web data. The measurement scale for chain length is links per chain. The measurement scale for inverse chain length is chains per link, or the *link fraction*. There is no reason to prefer links per chain to chains per link.

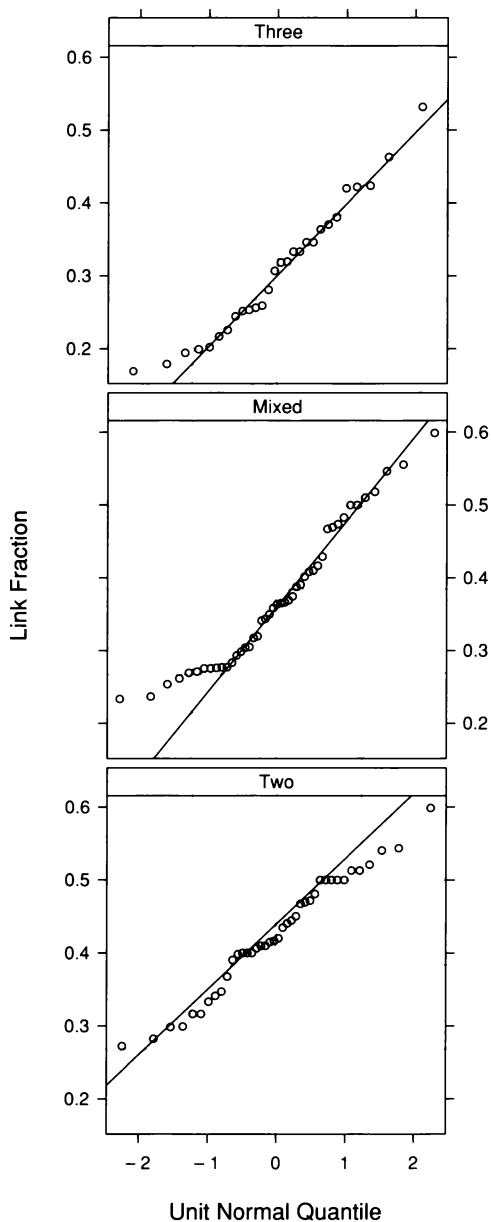
Fitting and Residuals

The normal q-q plots of the distributions in Figure 2.40 suggest that the shifts in the link-fraction distributions are additive on the inverse scale. Thus we will fit the data by estimating location, using means



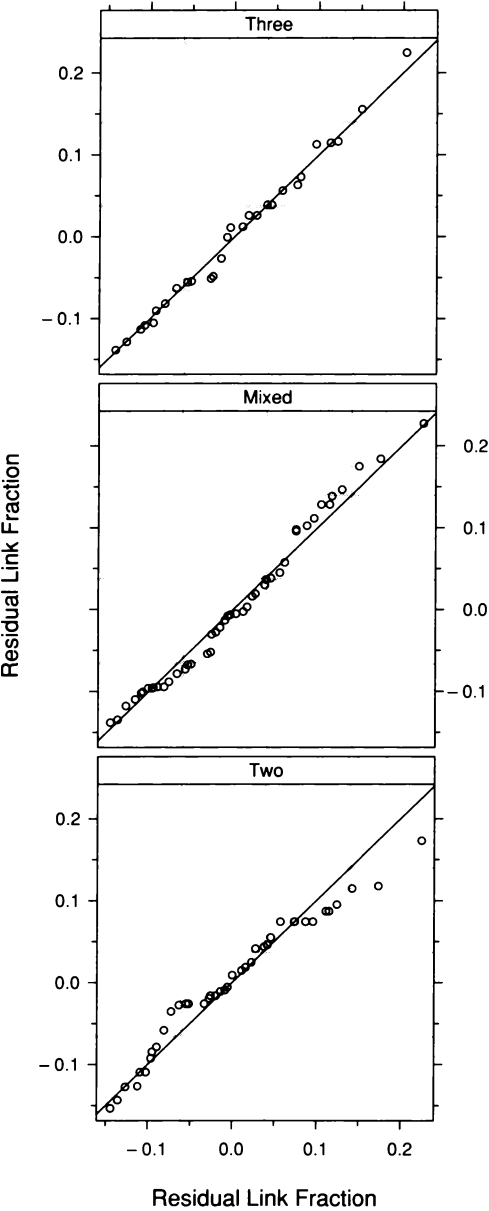
2.39 The s-l plot for the inverse chain lengths, or link fractions, checks for nonuniform spread.

because the distributions are not badly nonnormal. The mean link fractions are 0.43, 0.37, and 0.31 for two, mixed, and three. Note that these values are very close to $7/16$, $6/16$, and $5/16$. Noting this form is likely just pure numerology, a taking of pleasure from detecting a simple pattern, but without theoretical significance.



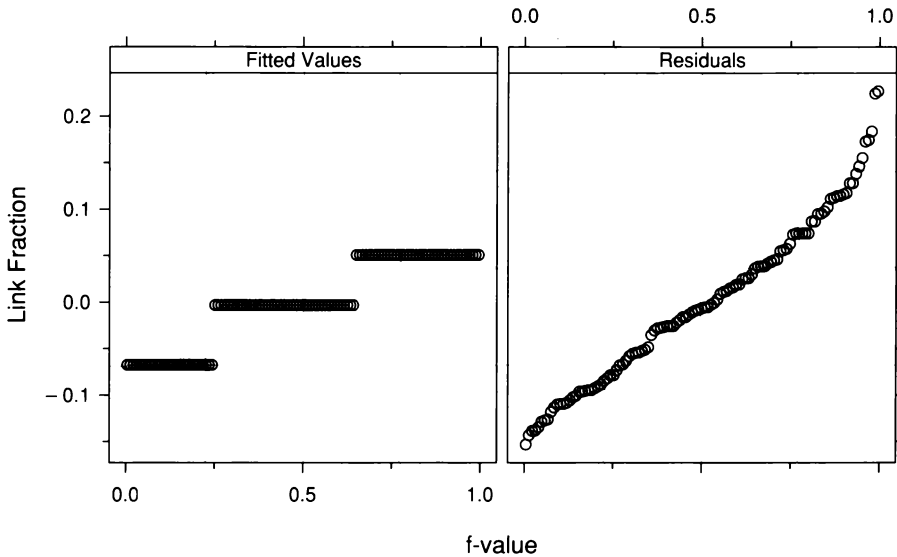
2.40 Normal q-q plots compare the normal distribution with the distributions of link fractions.

Figure 2.41 is a q-q plot of the residuals; the quantiles of each residual distribution are graphed against the quantiles of all residuals. The distributions are reasonably similar, so we can pool them.



2.41 Each panel is a q-q plot that compares the distribution of the residuals for one group of link fractions with the distribution of the pooled residuals.

Figure 2.42, an r-f spread plot, shows that the residual spread is considerably greater than the fitted-value spread. Thus, while the ecosystem dimension does affect the link fraction, it accounts for only a modest amount of the variation in the fractions.



2.42 An r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the fit to the link fractions.

The Force of Power Transformation

The amount of pushing and pulling on the tails of a distribution that is exerted by power transformations can be roughly measured by the ratio of the largest observation to the smallest. Data sets with large ratios are more sensitive to power transformation than data sets with small ratios. For the stereogram data, the ratio is 27.8 for the NV times and 19.7 for the VV times. For the food web data, it is 2.2 for two dimensions, 2.6 for mixed, and 3.1 for three. Power transformation clearly affects these two sets of data. If the ratio is too close to 1, power transformations with τ from -1 to 1 do not have much effect. For the eight singer distributions, the ratios range from 1.1 to 1.2, and power transformation has little effect on the shape of the distributions.

2.8 *Complex Shifts and Robust Fitting*

The visualization of the singer heights showed that the distributions differ by additive shifts, a simple structure that allows fitting by estimating location to produce homogeneous residuals. And the visualization showed that the residuals have a near-normal distribution. The discovery of power transformations for the fusion times and food web lengths resulted in measurement scales for which the distributions of these data sets also have additive shifts and near normality. Nature, however, is frequently not so obliging. Shifts between distributions can have an incurable complexity that makes them more difficult to characterize. And distributions can be severely nonnormal. Still, fitting and graphing can yield visualizations that provide a clear picture of the complexity and nonnormality.

Bin Packing Data

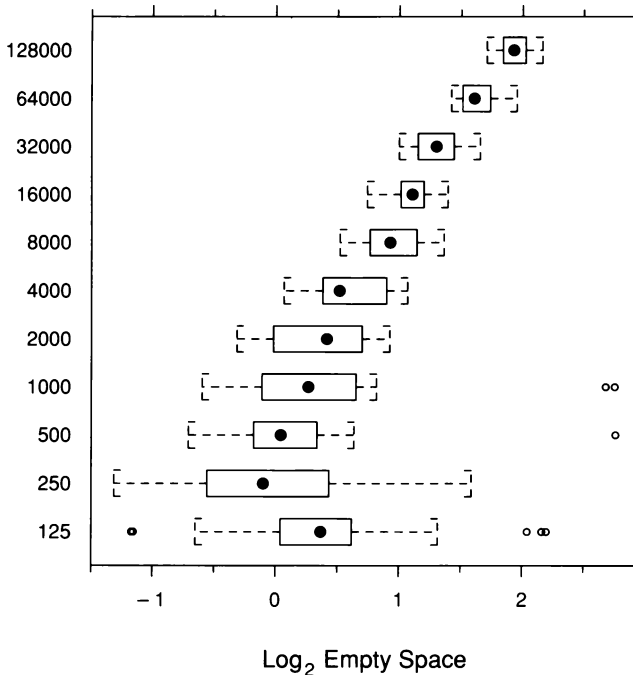
Bin packing is a computer problem that has challenged mathematicians working on the foundations of theoretical computer science. Suppose a large number of files of different sizes are to be written on floppies. No file can be split between two floppies, but we want to waste as little space as possible. Unfortunately, any algorithm that guarantees the minimum possible empty space takes an enormous amount of computation time unless the number of files is quite small. Fortunately, there are heuristic algorithms that run fast and do an extremely good job of packing, even though they do not guarantee the minimum of empty space. One is *first fit decreasing*. The files are packed from largest to smallest. For each file, the first floppy is tried; if it has sufficient empty space, the file is written, and if not, the second floppy is tried. If the second file has sufficient space, the file is written and if not, the third floppy is tried. The algorithm proceeds in this way until a floppy with space, possibly a completely empty one, is found.

To supplement the theory of bin packing with empirical results, mathematicians and computer scientists have run simulations, computer experiments in which bins are packed with randomly generated weights. For one data set from one experiment [8], the weights were randomly selected from the interval 0 to 0.8 and packed in bins of size one. The number of weights, n , for each simulation run took one

of 11 values: 125, 250, 500, and so forth by factors of 2 up to 128000. There were 25 runs for each of the 11 different numbers of weights, which makes $25 \times 11 = 275$ runs in all. For each run of the experiment, the performance of the algorithm was measured by the total amount of empty space in the bins that were used. We will study log empty space to enhance our understanding of multiplicative effects.

Shifts in Location, Spread, and Shape

Figure 2.43 displays the bin packing data by box plots. The shifts in the 11 distributions are complex. The locations tend to increase with n , the number of weights. For example, the medians increase with n except for the two smallest values of n . The box plots also show that the spreads of the distributions tend to decrease as n increases. Both the interquartile ranges and the ranges of the adjacent values tend to decrease with n .

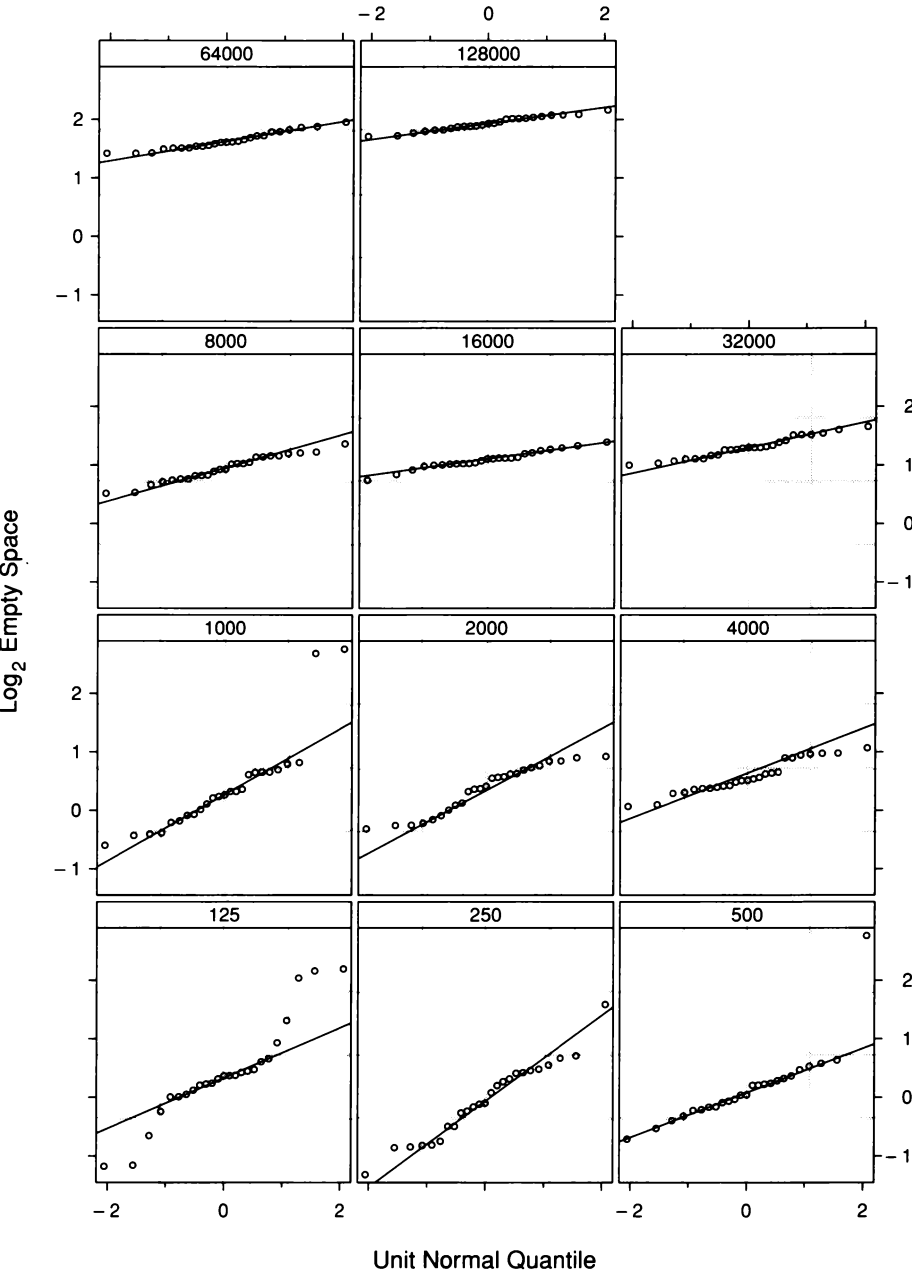


2.43 Box plots compare the distributions of the bin packing data.

Figure 2.44, normal q-q plots of the bin packing data, show that not all of the bin packing distributions are well approximated by the normal. For n equal to 2000 to 128000, the patterns are reasonably straight, but for lesser values of n , there are marked deviations from linear patterns. In other words, not only do the distributions differ in location and spread, they differ in shape as well. To study this shape in more detail, we will fit the data, compute residuals, and study the residual distributions. But since both location and spread change, we need to estimate both in fitting the data.

Figure 2.44 reveals enough information about shape to have a major impact on how we carry out the fitting of location and spread. For $n = 125$, the line follows the middle of the data; if the entire distribution were well approximated by the normal, the points in the tails would lie near this line. Instead, the points in the upper tail are above the line, and the points in the lower tail are below the line. In other words, with respect to a normal distribution as determined by the middle of the data, the values in the tails are too spread out. Distributions with this property occur frequently enough that there is a term to describe their shape — *leptokurtic*. The root “lepto” means slender. For data that are leptokurtic, the relative density of the data in the middle of the distribution compared with the density in the tails is less (thinner) than for a normal distribution. In cases where leptokurtosis affects only the extreme tails, the result can be just a few outliers. For n equal to 500 and 1000, three observations are considerably larger than the others. These outliers are likely just a remnant of a leptokurtosis that moves further and further into the tails of the distribution as n increases through 1000.

The opposite of leptokurtosis is *platykurtosis*. With respect to a normal distribution as determined by the middle of the data, the values in the tails of the data are a bit too close to the middle, that is, not sufficiently spread out. The root “platy” means broad. For data that are platykurtic, the relative density of the data in the middle of the distribution compared with the density in the tails is greater (broader) than for a normal distribution. Shortly, we will also see platykurtosis in the bin packing data.



2.44 Normal q-q plots compare the distributions of the bin packing data with the normal distribution.

Robust Estimation

For the singer heights, the log fusion times, and the link fractions, the distributions of the different groups of measurements are reasonably well approximated by the normal. For this reason, in fitting the data, we used sample means to estimate the locations of the distributions. Had it been necessary to account for changing spread, we would have used sample standard deviations to fit spread. Statistical theory tells us that sample means and standard deviations are the right thing to use when the distributions are close to the normal. And when a distribution is mildly platykurtic, sample means and standard deviations are typically satisfactory.

But in the presence of leptokurtosis, means and standard deviations can perform disastrously. Even just a few outliers can yield a silly answer. Suppose we have 20 observations ranging from 10 to 20 with a mean of 16.3. Suppose we add one new observation, an outlier equal to 99. The mean of the new data now becomes

$$(20 \times 16.3 + 99)/21 = 20.2 .$$

This is not a sensible description of the location of the data because it exceeds all of the observations except one. The sample standard deviation is also sensitive to a small number of outliers.

The median is typically *robust*: resistant to a few outliers or to more extreme leptokurtosis [51, 65]. Suppose that for the 20 observations above, the two middle observations — that is, the 10th and 11th largest observations — are 14 and 15. Then the median is 14.5. When the outlier is added, the median changes to 15, a sensible answer. Similarly, the mad, introduced earlier for use on s-l plots, is a robust estimate of spread [51, 65]. Thus we will use medians to fit the locations of the bin packing distributions, and mads to fit the spreads.

Let b_{in} be the i th log empty space measurement for the bin packing run with n weights. Let ℓ_n be the medians, and let s_n be the mads. The fitted values are

$$\hat{b}_{in} = \ell_n ,$$

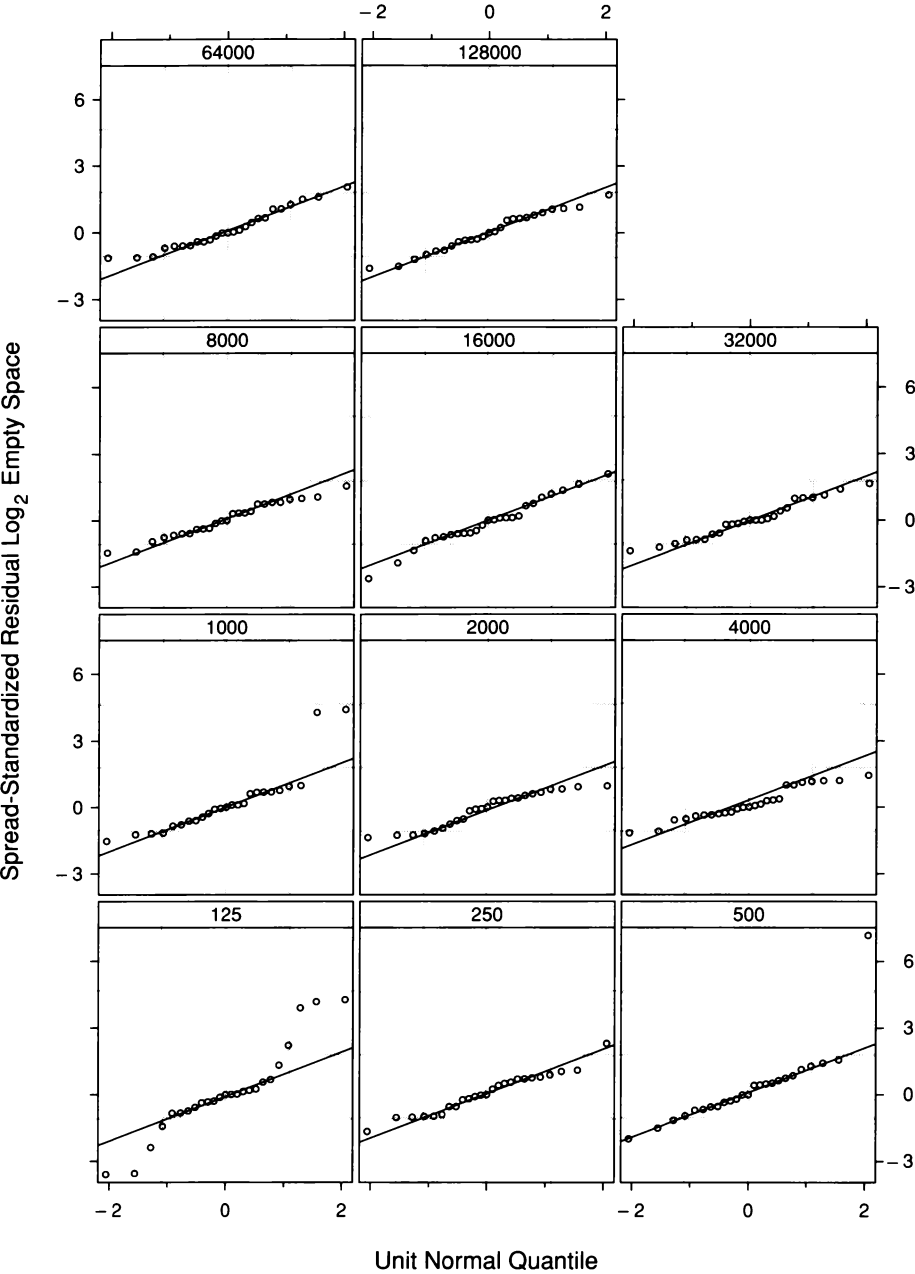
the residuals are

$$\hat{\varepsilon}_{in} = b_{in} - \hat{b}_{in} ,$$

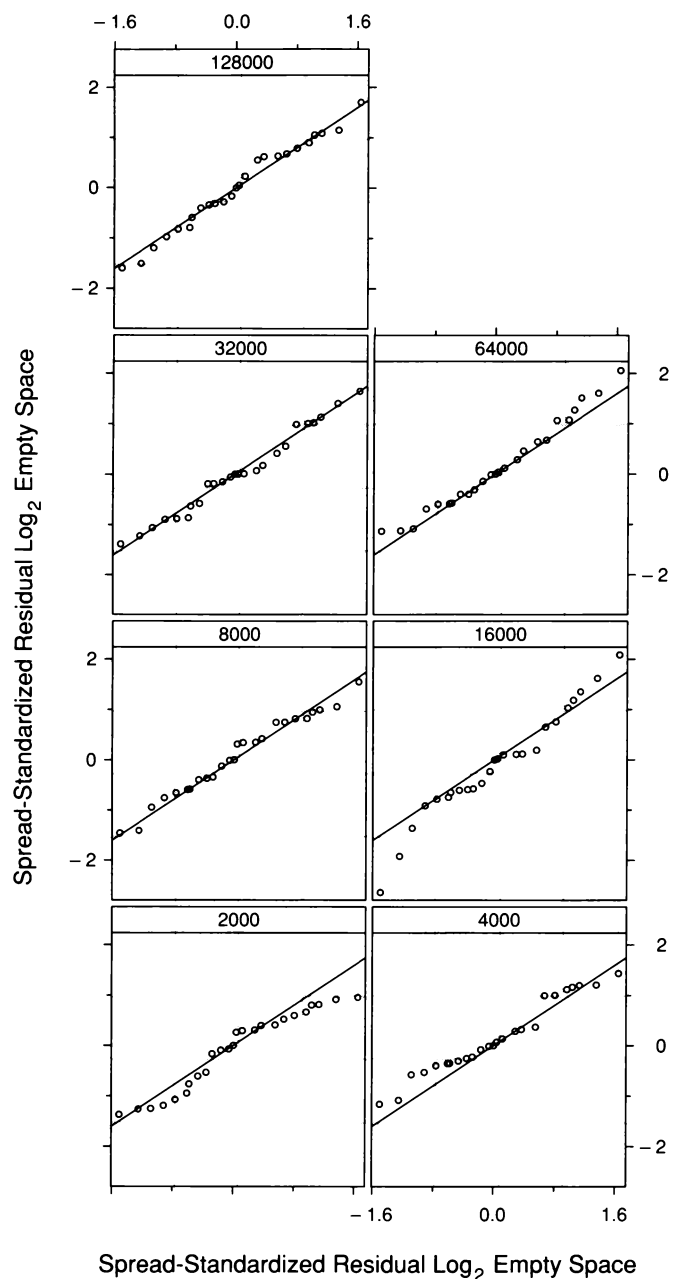
and the *spread-standardized residuals* are

$$\frac{\hat{\varepsilon}_i}{s_n} .$$

In Figure 2.45, the distributions of the 11 sets of spread-standardized residuals are graphed by normal q-q plots. As before, the leptokurtosis is clearly revealed, but now we also see that once n is above 1000, the distributions turn mildly platykurtic. But a few outliers for distributions with $n \leq 1000$ squash the spread-standardized residuals for $n > 1000$ into a small region of the vertical scale, which interferes with our assessment of the platykurtosis. To enhance the assessment, we will eliminate the four distributions with $n \leq 1000$, and analyze the remaining seven residual distributions. Figure 2.46 shows q-q plots of each residual distribution against the pooled distribution of the seven sets. The departures for the three smaller values of n appear somewhat bigger than for the remaining, so there are differences among the distributions, but the effect is slight.



2.45 Normal q-q plots display spread-standardized residuals from the robust fit to the bin packing data.



2.46 The q-q plots compare distributions of spread-standardized residuals for seven values of n with the pooled values of the seven distributions.